Math 371 Exam 2 Name Darrell Long Page 1

96/100

1. Give an example to show that even if two relations R and S are both transitive over the set X, R ∪ S may still not be transitive.

5

Let $R_1 = \{\langle 1,2\rangle, \langle 2,3\rangle, \langle 1,3\rangle\}$

$S_2 = \{\langle 3,5\rangle\}$

$R \cup S = \{\langle 1,2\rangle, \langle 2,3\rangle, \langle 1,3\rangle, \langle 3,5\rangle\}$

could have used R : <
S : >
as like first guess.

2. Consider the relation R in X, where X is the set of <u>all</u> integers, and R = {<x,y> : x-y is a multiple of three}

a) Prove that R is an equivalence relation

20

1) for any x, $x - x = \emptyset$ and $\emptyset$ is a multiple of 3; R is reflexive

2) for any x and y, and α is a multple of 3.

if $x - y = \alpha$, then $x = \alpha + y$

$x - \alpha = y$

$\boxed{-\alpha = y - x;}$

and since if α is a multiple of three, the $-\alpha$ is a mutiple of 3, hence R is symetric

3) for any x, y and z, and α is a multiple of 3. and m, n, o are arbitrary integers

if $x - y = m\alpha$ and $y - z = n\alpha$ then $x - z = o\alpha$

$x - y = m\alpha$

$y - z = n\alpha$

$\boxed{x - z = (m+n)\alpha}$ and since if α is a multiple of 3, $k\alpha$ is too, then R is transitive

b) List all the distinct equivalence classes

-3

$[0]_R = \{-3, -6, -9, \ldots\}$

$[3]_R = \{0, 3, 6, 9, \ldots\}$

$[0]_R = \{\ldots, -6, -3, \ldots, 6 \ldots,\}$

$[1]_R = \{\ldots, -5, -2, 1, 4, 7, \ldots\}$

$[2]_R = \{\ldots, -4, -1, 2, 5, 8, \ldots\}$

3. Use matrix methods to determine whether the relation represented by

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

is reflexive: yes (1)
is symmetric: no (2)
is irreflexive: no (1)
is antisymmetric: no (3)
is transitive: yes (1)
(Justify your answers)

-1

15

1) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ reflexive, not irreflexive

2) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ \boxed{1} & 0 & 1 \end{bmatrix} \neq \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ not symetric

?

3) $\begin{bmatrix} 1\wedge1 & 1\wedge0 & 1\wedge1 \\ 0\wedge1 & 1\wedge1 & 0\wedge0 \\ 1\wedge0 & 0\wedge0 & 1\wedge1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ not antisemetric

4) $M_1^2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ $M_1 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

transitive

4. Show that the conclusion C: $(\neg(\exists x)\ P(x)) \wedge (\exists z)\neg Q(z)$ can be deduced from the hypothesis H: $\neg(\exists y)(P(y) \vee Q(y))$.

Identify each implication that you use.

H: $\neg(\exists y)(P(y) \vee Q(y))$ C: $(\neg(\exists x)P(x)) \wedge (\exists z)\neg Q(z)$

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1)	$\neg(\exists y)(P(y) \vee Q(y))$	P
2)	$(\forall y)\neg(P(y) \vee Q(y))$	E_{25} (1)
3)	$\neg(P(a) \vee Q(a))$	US (2)
4)	$\neg P(a) \wedge \neg Q(a)$	E_9 (3)
5)	$\neg P(a)$	I_1 (4)
6)	$\neg Q(a)$	I_2 (4)
7)	$(\exists z)\neg Q(z)$	EG (6)
8)	$(\forall x)\neg P(x)$	UG (5) — nothing special about a
9)	$\neg(\exists x)P(x)$	E_{25} (8)
10)	$(\neg(\exists x)P(x)) \wedge (\exists z)\neg Q(z)$	I_9 (7, 9)

5. Prove that if two relations R and S (in X) are both reflexive, then $R \cup S$ is also reflexive.

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1)	$(\forall x \in X)(\langle x,x\rangle \in R)$	P (R reflex)
2)	$(\forall x \in X)(\langle x,x\rangle \in S)$	P (S reflex) — not needed
3)	$\langle y,y\rangle \in R$	US
4)	$\langle y,y\rangle \in R \vee \langle y,y\rangle \in S$	I_3 (3)
5)	$\langle y,y\rangle \in (R \cup S)$	def^{n} of $\cup$
6)	$(\forall x \in X)(\langle x,x\rangle \in (R \cup S))$	UG (5) (R∪S Reflex)

6. Rigorously prove that C: $A \subseteq D$ follows from the hypotheses H_1: $A \subseteq B$ H_2: $B \subseteq D$.

Identify each implication that you use.

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C: $A \subseteq D \Leftrightarrow (\forall x)(x \in A \rightarrow x \in D)$ H_1: $A \subseteq B \Leftrightarrow (\forall x)(x \in A \rightarrow x \in B)$
H_2: $B \subseteq D \Leftrightarrow (\forall x)(x \in B \rightarrow x \in D)$

1)	$(\forall x)(x \in A \rightarrow x \in B)$	P
2)	$y \in A \rightarrow y \in B$	US (1)
3)	$(\forall x)(x \in B \rightarrow x \in D)$	P
4)	$y \in B \rightarrow y \in D$	US (3)
5)	$y \in A \rightarrow y \in D$	I_{13} (2, 4)
6)	$(\forall x)(x \in A \rightarrow x \in D)$	UG (5)

QED